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Implement a Stack data structure in Python with the following methods:

- push(item): add item to top
- pop(): remove and return top item
- peek(): return top without removing
- is_empty(): return True if stack is empty

Page 2 - Requirements

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Input Format:

- Commands come from stdin, one per line
- Commands: PUSH <value>, POP, PEEK, EMPTY

Output Format:

- POP prints the popped value or "Stack empty"
- PEEK prints the top value or "Stack empty"
- EMPTY prints "True" or "False"

Constraints:

- Use a list internally (do NOT use `collections.deque` for pop)
- Handle empty stack gracefully
- Maximum 10,000 operations

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Create a file main.py with:

```
class Stack:  
    def __init__(self):  
        self._items = []
```

```
def push(self, item):  
    pass # TODO
```



```
def pop(self):  
    pass # TODO
```

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Input:

PUSH 5

PUSH 3

PEEK

POP

EMPTY

Expected Output:

3

3

False

Page 5 - Testing

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Run tests with: `pytest test_main.py`

All tests must pass before submission.